# Mathematics for Computer Science

Mihir Rao

## Contents

**1 Predicates, Sets, and Proofs** **4**
1.1 Propositions and mathematical proof . . . 4
1.2 Logical connectives . . . 4
1.3 Sets and tuples . . . 4
1.4 Axioms, consistency, and completeness . . . 5

**2 Contradiction and Induction** **6**
2.1 Deduction and proof structure . . . 6
2.2 Contradiction . . . 6
2.3 Induction . . . 6
2.4 Strengthening the hypothesis . . . 7

**3 Casework and Strong Induction** **8**
3.1 Exhaustive cases . . . 8
3.2 Strong induction . . . 8
3.3 Splitting stacks . . . 9
3.4 An ordering of tournament players . . . 9

**4 State Machines** **10**
4.1 Executions and invariants . . . 10
4.2 The eight puzzle . . . 10
4.3 Potential functions and termination . . . 11
4.4 Adjacent-swap sorting . . . 11

**5 Sums** **12**
5.1 Geometric sums and discounted payments . . . 12
5.2 Perturbation, guessing, and double sums . . . 12
5.3 Integral bounds . . . 13
5.4 The reciprocal-square sum . . . 13

**6 Asymptotics** **14**
6.1 Overhanging blocks and harmonic numbers . . . 14
6.2 Products and factorials . . . 14
6.3 Asymptotic comparisons . . . 15
6.4 Using the notation correctly . . . 15

**7 Recurrences** **16**
7.1 Recursive descriptions . . . 16
7.2 Towers of Hanoi . . . 16
7.3 Sorting recurrences . . . 16
7.4 The master theorem . . . 17

**8 Divisibility** **18**
8.1 Divisors and integer linear combinations . . . 18
8.2 Greatest common divisors and remainders . . . 18
8.3 Euclid's algorithm . . . 18
8.4 The extended algorithm and Bézout's identity . . . 19

**9 Modular Arithmetic** **20**
9.1 Congruence and residues . . . 20
9.2 Arithmetic and exponentiation . . . 20
9.3 Inverses and cancellation . . . 20
9.4 Fermat's little theorem . . . 21
9.5 Digit tests and checksums . . . 21

**10 Cryptography** **22**
10.1 Encryption and secret keys . . . 22
10.2 One-time pads . . . 22
10.3 Diffie–Hellman key agreement . . . 22
10.4 RSA . . . 23
10.5 Chinese remainder theorem . . . 23

**11 Graphs and Coloring** **24**
11.1 Degrees and bipartitions . . . 24
11.2 Coloring as a conflict model . . . 24
11.3 Greedy coloring and its limits . . . 25

**12 Matching** **26**
12.1 Size, maximality, and weights . . . 26
12.2 Stable matching . . . 26
12.3 Gale–Shapley deferred acceptance . . . 26
12.4 Which stable matching is selected? . . . 27

**13 Connectivity and Trees** **28**
13.1 Walks, paths, and components . . . 28
13.2 Euler trails and tours . . . 28
13.3 Trees and leaf removal . . . 29

**14 Digraphs and DAGs** **30**
14.1 Direction and reachability . . . 30
14.2 Topological orders . . . 30
14.3 Parallel scheduling, chains, and antichains . . . 31

**15 Relations and Counting** **32**
15.1 Relations, functions, and set sizes . . . 32
15.2 Equivalence and order . . . 32
15.3 Basic counting rules . . . 33

**16 Counting Techniques** **34**
16.1 Division and combinations . . . 34
16.2 Poker-hand recipes . . . 34

**17 More Counting Techniques** **36**
17.1 Inclusion–exclusion . . . 36
17.2 Pigeonhole arguments . . . 36
17.3 Combinatorial proofs and expansions . . . 37

**18 Introduction to Probability** **38**
18.1 Discrete spaces and the tree method . . . 38
18.2 Monty Hall . . . 38
18.3 Intransitive dice . . . 39

**19 Conditional Probability** **40**
19.1 Rules for events . . . 40
19.2 Conditioning and products . . . 40
19.3 Bayes' rule and base rates . . . 41
19.4 Aggregating groups and conditioning on evidence . . . 41

**20 Independence** **42**
20.1 Pairwise and mutual independence . . . 42
20.2 Conditional independence . . . 43
20.3 Birthday collisions . . . 43
20.4 A known model versus an unknown coin . . . 43

**21 Random Variables** **44**
21.1 Functions on outcomes . . . 44
21.2 Distributions . . . 44
21.3 A randomized envelope strategy . . . 44
21.4 The binomial distribution . . . 45

**22 Expectation** **46**
22.1 Weighted averages and tail sums . . . 46
22.2 Linearity without independence . . . 46
22.3 Randomly returned phones . . . 47
22.4 Checking fairness by expectation . . . 47

**23 Expectation and Variance** **48**
23.1 How many events occur? . . . 48
23.2 Products and ratios . . . 48
23.3 Variance and moments . . . 49
23.4 Computing and combining variances . . . 49

**24 Large Deviations** **51**
24.1 Markov's inequality . . . 51
24.2 Chebyshev's inequality . . . 51
24.3 Chernoff's bound . . . 52

# Lecture 1: Predicates, Sets, and Proofs

## 1.1 Propositions and mathematical proof

A proposition is a statement with a definite truth value. A predicate is a statement whose truth depends on its variables. For example, $n^2 + n + 41$ is prime is a predicate on $n \in \mathbb{N}$, whereas $\forall n \in \mathbb{N},\ n^2 + n + 41$ is prime is a proposition. Here $\mathbb{N} = \{0, 1, 2, \ldots\}$; $\forall$ means for every, and $\exists$ means there exists.

A proof verifies a proposition by logical deductions from specified axioms, statements assumed true. Experiments and examples can suggest a conjecture but cannot establish a universal claim. The polynomial above produces primes for $n = 0, \ldots, 39$, yet at $n = 41$ its value is $41 \cdot 43$. Large counterexamples can also refute seemingly convincing patterns. For instance, $95800^4 + 217519^4 + 414560^4 = 422481^4$ contradicts the claim that a fourth power cannot be the sum of three positive fourth powers.

## 1.2 Logical connectives

For propositions $P, Q$, write $\neg P$ for not $P$, $P \wedge Q$ for both, and $P \vee Q$ for at least one. Mathematical or is inclusive.

| $P$ | $Q$ | $P \wedge Q$ | $P \vee Q$ | $P \Rightarrow Q$ | $P \Leftrightarrow Q$ |
|---|---|---|---|---|---|
| $T$ | $T$ | $T$ | $T$ | $T$ | $T$ |
| $T$ | $F$ | $F$ | $T$ | $F$ | $F$ |
| $F$ | $T$ | $F$ | $T$ | $T$ | $F$ |
| $F$ | $F$ | $F$ | $F$ | $T$ | $T$ |

An implication $P \Rightarrow Q$ excludes exactly the case where $P$ is true and $Q$ is false. It does not assert causation or a temporal relationship. Its contrapositive $\neg Q \Rightarrow \neg P$ is equivalent; its converse $Q \Rightarrow P$ generally is not. A biconditional $P \Leftrightarrow Q$ asserts both directions. The stronger condition $x < 3$ implies $x < 4$, although the reverse fails.

## 1.3 Sets and tuples

A set is an unordered collection without repetitions. Membership is written $x \in A$, and $A \subseteq B$ means every member of $A$ belongs to $B$. The empty set is $\varnothing$. Set-builder notation specifies a set by a predicate, as in $\{n \in \mathbb{N} : n \text{ is prime}\}$. Basic operations are

$$A \cup B = \{x : x \in A \text{ or } x \in B\}, \quad A \cap B = \{x : x \in A \text{ and } x \in B\}, \quad A \setminus B = \{x : x \in A,\ x \notin B\}.$$

Sets may themselves be elements: $\{2, \{3, 4\}, \varnothing\}$ has three elements. An ORDERED TUPLE, such as $(6, 1, 2, 0)$, retains order and permits repetition; it differs from $(2, 1, 6, 0)$.

### 1.4 Axioms, consistency, and completeness

Different axiom systems describe different mathematical structures. For example, alternative assumptions about parallels through a point outside a line lead to different geometries. An axiom system is CONSISTENT if it cannot prove both a proposition and its negation, and COMPLETE if it decides every proposition in its language. Gödel's incompleteness theorem says that a consistent, effectively axiomatized system capable of expressing sufficient arithmetic cannot be complete. The arithmetic and effectiveness assumptions matter; the statement does not apply to every possible axiom system.

**Source.** Z. Abel, B. Chapman, and E. Demaine, *Mathematics for Computer Science*, Lecture 1 (MIT 6.1200J, Spring 2024). Adapted under CC BY-NC-SA 4.0.

# Lecture 2: Contradiction and Induction

## 2.1 Deduction and proof structure

An inference rule combines established propositions into a new one. Modus ponens derives $Q$ from $P$ and $P \Rightarrow Q$; modus tollens derives $\neg P$ from $P \Rightarrow Q$ and $\neg Q$. Implications compose: $P \Rightarrow Q$ and $Q \Rightarrow R$ yield $P \Rightarrow R$.

The logical form of a goal suggests its proof structure:

- To prove $\exists x \in S,\ P(x)$, construct a witness and verify it. The integer 17 witnesses the existence of a prime at least 10.
- To prove $\forall x \in S,\ P(x)$, fix an arbitrary $x \in S$ and derive $P(x)$ without extra assumptions. For every real $x$, $x^2 - 6x = (x-3)^2 - 9 \geq -9 > -10$.
- To prove $P \Rightarrow Q$, assume $P$ and establish $Q$, or prove its contrapositive. If $n$ is odd, $n = 2k+1$ gives $n^2 = 2(2k^2 + 2k) + 1$, proving that an even square has an even root.
- To prove an equivalence, prove both implications. Nested quantifiers and implications determine which variables may be fixed and which hypotheses are available at each step.

## 2.2 Contradiction

To prove $P$ by contradiction, assume $\neg P$ and derive a false statement.

**Theorem 2.1.** *The number $\sqrt{2}$ is irrational.*

*Proof.* Suppose $\sqrt{2} = a/b$ with integers $a, b$, $b \neq 0$, in lowest terms. Then $a^2 = 2b^2$, so $a$ is even. Writing $a = 2c$ gives $b^2 = 2c^2$, so $b$ is even too. This contradicts the choice of lowest terms. □

## 2.3 Induction

The induction principle proves $P(n)$ for all $n \in \mathbb{N}$ from a base case $P(0)$ and an inductive step $\forall n \geq 0,\ P(n) \Rightarrow P(n+1)$. The step assumes one instance as a temporary hypothesis; it does not assume the entire theorem.

**Proposition 2.2.** *For every $n \in \mathbb{N}$, $\sum_{i=1}^{n} i = n(n+1)/2$.*

*Proof.* At $n = 0$, both sides are zero. If the identity holds at $n$, then

$$\sum_{i=1}^{n+1} i = \frac{n(n+1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2}.$$

This proves the step and hence the identity. □

## 2.4 Strengthening the hypothesis

Every $2^n \times 2^n$ board with any one square removed can be tiled by L-shaped trominoes. This stronger statement is easier to induct on than the special case where the removed square is near the center.

*Proof.* For $n = 0$ there is nothing to tile. Divide a board of side $2^{n+1}$ into four quadrants. Place one central tromino covering a central square in each of the three quadrants that do not contain the missing square. Each quadrant is now a board of side $2^n$ with one square already missing or occupied. Apply the induction hypothesis to each quadrant. □

The argument constructs a tiling recursively. Strengthening the statement adds an obligation to the conclusion, but also makes the induction hypothesis useful for all four subproblems.

**Source.** Z. Abel, B. Chapman, and E. Demaine, *Mathematics for Computer Science*, Lecture 2 (MIT 6.1200J, Spring 2024). Adapted under CC BY-NC-SA 4.0.

# Lecture 3: Casework and Strong Induction

## 3.1 Exhaustive cases

To prove $P$, it suffices to prove it under each of conditions $C_1, \ldots, C_k$ whose disjunction is always true. Cases need not be disjoint, but they must cover every possibility. For example, $(A \Rightarrow B) \vee (B \Rightarrow C)$ is a tautology: when $B$ is true the first implication holds, and when $B$ is false the second holds.

**Theorem 3.1.** *Among six people there are three mutual friends or three mutual strangers.*

*Proof.* Fix a person $p$. Among the other five, at least three are friends of $p$ or at least three are strangers to $p$. In the first case, choose three friends. If some pair are friends, they and $p$ give a friend triple; otherwise the chosen three are mutual strangers. The second case follows by interchanging friendship and strangerhood. □

The Ramsey number $R(f, s)$ is the least group size forcing $f$ mutual friends or $s$ mutual strangers. Thus $R(3, 3) \leq 6$; in fact $R(3, 3) = 6$ and $R(4, 4) = 18$. Exhaustive case arguments may be large: the four-color theorem asserts that four colors suffice for a planar map, with regions sharing a boundary segment receiving different colors. Its proof requires much more than checking sample maps.

## 3.2 Strong induction

Strong induction permits assuming $P(0), \ldots, P(n)$ when proving $P(n + 1)$. Together with the base case this proves $P(n)$ for all natural $n$. It is equivalent to ordinary induction: apply ordinary induction to the predicate $Q(n) = \bigwedge_{i=0}^{n} P(i)$. The extra hypotheses are useful when a problem splits into pieces of different sizes.

### 3.3 Splitting stacks

Start with $n$ blocks. Splitting a stack into positive sizes $a, b$ earns $ab$ points. Continue until every stack has size one.

**Theorem 3.2.** *Every strategy earns exactly $n(n-1)/2$ points.*

*Proof.* Use strong induction on $n \geq 1$. A singleton earns zero. For $n > 1$, the first split has $a + b = n$ and $1 \leq a, b < n$. Later splits of the two piles are independent, so the induction hypothesis gives total

$$ab + \frac{a(a-1)}{2} + \frac{b(b-1)}{2} = \frac{(a+b)^2 - (a+b)}{2} = \frac{n(n-1)}{2}.$$

□

In particular, splitting off one block at a time earns $1 + 2 + \cdots + (n-1)$, which therefore equals every other strategy's score.

### 3.4 An ordering of tournament players

A TOURNAMENT has one directed outcome for each pair of players, with no ties. A BEATS ORDERING lists players $p_1, \ldots, p_n$ so that $p_i$ beat $p_{i+1}$ for each $i < n$; it need not order every pair by who won.

**Theorem 3.3.** *Every tournament has a beats ordering.*

*Proof.* Use strong induction, including the empty tournament as base case. Choose a player $p$ and partition the others into $W$, those who beat $p$, and $L$, those whom $p$ beat. Each smaller induced tournament has a beats ordering by induction. Concatenate the ordering of $W$, then $p$, then the ordering of $L$. The only new consecutive pairs involve $p$, and their orientations follow from the definitions of $W, L$. Empty parts cause no difficulty. □

**Source.** Z. Abel, B. Chapman, and E. Demaine, *Mathematics for Computer Science*, Lecture 3 (MIT 6.1200J, Spring 2024). Adapted under CC BY-NC-SA 4.0.

# Lecture 4: State Machines

## 4.1 Executions and invariants

A state machine consists of states, an initial state, and allowed transitions $s \to t$. It is deterministic when each state has at most one successor, and nondeterministic otherwise. An execution follows the transitions from the initial state; a state is reachable if some execution visits it.

A predicate is preserved when $P(s)$ and $s \to t$ imply $P(t)$. An invariant holds at every reachable state.

**Theorem 4.1 [Invariant principle].** *A preserved predicate true initially is an invariant.*

*Proof.* Induct on the number of transitions. The base is the initial state. If the predicate holds after $k$ transitions, preservation gives it after the next transition. Thus it holds along every execution. □

A state violating an invariant is unreachable. Finding a useful invariant is the substantive part of applying the principle.

## 4.2 The eight puzzle

In the eight puzzle a tile adjacent to the blank may slide into it. Compare the initial and target boards

$$\begin{array}{ccc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 8 & 7 & * \end{array} \qquad \longrightarrow \qquad \begin{array}{ccc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & * \end{array}.$$

Read a board row by row and delete the blank. An inversion is a pair $i < j$ with $a_i > a_j$. The initial sequence has one inversion; the target has zero.

A horizontal slide leaves the sequence unchanged. A vertical slide moves one entry past exactly two others. Each crossing toggles one pair's inversion status, so the inversion count changes by an even integer. Therefore odd inversion parity is preserved and holds initially. The target's even parity proves that it is unreachable. The exact inversion count is not preserved: a vertical move can change it by $-2$, $0$, or $2$.

## 4.3 Potential functions and termination

A FINAL STATE has no outgoing transitions. A machine TERMINATES if every execution is finite. A POTENTIAL FUNCTION, or derived variable, assigns a number to each state.

**Theorem 4.2.** *If a natural-number-valued potential strictly decreases at every transition, the machine terminates.*

*Proof.* An infinite execution would give an infinite strictly decreasing sequence of natural numbers, which is impossible. □

Strict decrease of real numbers alone is insufficient; natural-number values provide a bound on how many decreases can occur.

## 4.4 Adjacent-swap sorting

Given a sequence, repeatedly swap any adjacent pair that is out of order. States are permutations of the input; transitions are the allowed swaps. A final state has $a_i \leq a_{i+1}$ for every $i$, and hence is sorted by transitivity.

Use the inversion count as a potential. Swapping an inverted adjacent pair removes their inversion, while comparisons involving every other entry contribute the same total as before. Thus each move decreases the potential by exactly one. The machine terminates at a sorted sequence after exactly the initial number of inversions, at most $\binom{n}{2}$. The final-state argument proves partial correctness; termination completes the correctness proof.

**Source.** Z. Abel, B. Chapman, and E. Demaine, *Mathematics for Computer Science*, Lecture 4 (MIT 6.1200J, Spring 2024). Adapted under CC BY-NC-SA 4.0.

# Lecture 5: Sums

## 5.1 Geometric sums and discounted payments

At a fixed annual interest rate $p > 0$, a payment of $m$ after $k$ years has present value $m/(1+p)^k$. An annuity paying at times $0, 1, \ldots, n-1$ therefore has value

$$V = m \sum_{k=0}^{n-1} x^k = m \frac{1 - x^n}{1 - x}, \qquad x = (1+p)^{-1}.$$

This is a geometric sum. To derive it, set $S = 1 + x + \cdots + x^{n-1}$ and subtract $xS$: $(1-x)S = 1 - x^n$. For $x = 1$, the sum is $n$. For $|x| < 1$, taking a limit gives $\sum_{k=0}^{\infty} x^k = 1/(1-x)$. Thus an indefinite stream of payments can have finite present value. For the illustrative values $m = 50{,}000$, $n = 20$, $p = 0.0533$, the finite annuity is worth about 638,340; a perpetual annual unit payment starting now is worth $(1+p)/p \approx 19.76$.

## 5.2 Perturbation, guessing, and double sums

The perturbation method compares a sum with a shifted, scaled, or reversed copy to create cancellation. Another method is to guess a form and solve for unknown coefficients. For $S_n = \sum_{k=1}^{n} k^2$, a cubic guess $an^3 + bn^2 + cn + d$ and the values $n = 0, 1, 2, 3$ suggest

$$S_n = \frac{n(n+1)(2n+1)}{6}.$$

Matching finitely many values only suggests the formula. Verify it for all $n$ by checking $S_0 = 0$ and $S_{n+1} - S_n = (n+1)^2$. An empty sum has value zero.

For a double sum, identify its index region before changing order:

$$\sum_{i=1}^{n} \sum_{j=1}^{i} j = \frac{1}{2} \sum_{i=1}^{n} (i^2 + i) = \frac{n(n+1)(n+2)}{6},$$

$$\sum_{i=1}^{n} \sum_{j=i}^{n} j = \sum_{j=1}^{n} \sum_{i=1}^{j} j = \sum_{j=1}^{n} j^2.$$

Introducing a second sum can help too:

$$\sum_{j=1}^{n} j 2^j = \sum_{i=1}^{n} \sum_{j=i}^{n} 2^j = \sum_{i=1}^{n} (2^{n+1} - 2^i) = (n-1)2^{n+1} + 2.$$

## 5.3 Integral bounds

Put $I = \int_1^n f(x)\,dx$ and $S = \sum_{k=1}^n f(k)$.

**Theorem 5.1 [Integral comparison].** *If $f$ is weakly increasing, $f(1) + I \le S \le f(n) + I$. If $f$ is weakly decreasing, $f(n) + I \le S \le f(1) + I$.*

*Proof.* For an increasing function, on each $[k, k+1]$ the integral lies between $f(k)$ and $f(k+1)$. Sum over $k = 1, \dots, n-1$ and rearrange. Reverse the inequalities for a decreasing function. □

For instance, with $f(x) = \sqrt{x}$,

$$1 + \frac{2}{3}(n^{3/2} - 1) \le \sum_{k=1}^{n} \sqrt{k} \le \sqrt{n} + \frac{2}{3}(n^{3/2} - 1).$$

For a nonnegative decreasing $f$, the infinite series and improper integral converge together, and

$$\int_1^\infty f(x)\,dx \le \sum_{k=1}^{\infty} f(k) \le f(1) + \int_1^\infty f(x)\,dx.$$

Tail estimates sharpen bounds after computing initial terms. For example, $\sum_{k=4}^{\infty} k^{-2} \in [1/4, 5/16]$, so $\sum_{k=1}^{\infty} k^{-2} \in [232/144, 241/144]$.

## 5.4 The reciprocal-square sum

The exact value is $\sum_{k=1}^{\infty} k^{-2} = \pi^2/6$. Here is the trigonometric argument. Let $n = 2m+1$ and expand

$$\frac{\sin(nx)}{\sin^n x} = \operatorname{Im}(\cot x + i)^n = nP(\cot^2 x).$$

The monic degree-$m$ polynomial $P$ has coefficient $-\binom{n}{3}/n$ on $t^{m-1}$ and roots $\cot^2(k\pi/n)$ for $1 \le k \le m$. Hence

$$\sum_{k=1}^{m} \cot^2 \frac{k\pi}{n} = \frac{1}{n}\binom{n}{3} = \frac{(n-1)(n-2)}{6}.$$

For $0 < x < \pi/2$, $\sin x < x < \tan x$ gives $\cot x < 1/x < \csc x$. Square, sum, and use $\csc^2 x = 1 + \cot^2 x$ to obtain

$$A_n < \sum_{k=1}^{m} \frac{1}{k^2} < A_n + \frac{\pi^2 m}{n^2}, \qquad A_n = \frac{\pi^2(n-1)(n-2)}{6n^2}.$$

As $m \to \infty$, both bounds tend to $\pi^2/6$.

**Source.** Z. Abel, B. Chapman, and E. Demaine, *Mathematics for Computer Science*, Lecture 5 (MIT 6.1200J, Spring 2024). Adapted under CC BY-NC-SA 4.0.

# Lecture 6: Asymptotics

## 6.1 Overhanging blocks and harmonic numbers

Stack identical unit-width blocks, with each supported by the next. For stability, the center of mass of every upper stack must lie above its support. Number blocks from the top starting at zero, and measure $d_k$ leftward from the top block's right edge to the next support's right edge, with $d_0 = 0$. Putting each upper stack's center of mass at its support edge gives

$$kd_k = \frac{k}{2} + \sum_{i=0}^{k-1} d_i.$$

Subtracting the equation for $k-1$ yields $d_k = d_{k-1} + 1/(2k)$. Thus the top block overhangs the table by

$$d_n = \frac{1}{2}H_n, \qquad H_n = \sum_{k=1}^{n} \frac{1}{k},$$

where $H_n$ is the $n$th harmonic number. Four blocks give $H_4/2 = 25/24 > 1$ widths of overhang.

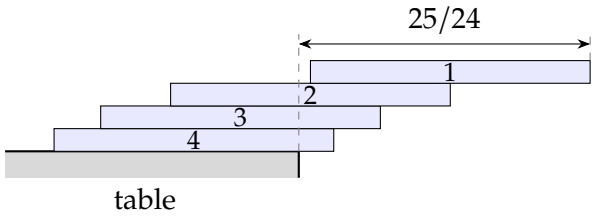



**Figure 1:** Four unit-width blocks at limiting equilibrium. Each upper stack's center of mass is at the right edge of its support.

Integral comparison gives $\ln n + 1/n \le H_n \le 1 + \ln n$, so $H_n \sim \ln n$: their ratio tends to one. In the ideal model, arbitrarily large overhang is possible, but an overhang of $D$ widths requires a number of blocks growing on the scale of $e^{2D}$.

## 6.2 Products and factorials

Taking logarithms converts products into sums: $\ln(n!) = \sum_{k=1}^{n} \ln k$. Since $\int_1^n \ln x \, dx = n \ln n - n + 1$, integral comparison and exponentiation give

$$\frac{n^n}{e^{n-1}} \le n! \le \frac{n^{n+1}}{e^{n-1}}.$$

The sharper Stirling formula is

$$n! = \sqrt{2\pi n}\left(\frac{n}{e}\right)^n e^{\epsilon_n}, \qquad \frac{1}{12n+1} < \epsilon_n < \frac{1}{12n}.$$

In particular $n! \sim \sqrt{2\pi n}(n/e)^n$. An additive error in the logarithm becomes a multiplicative error after exponentiation.

### 6.3 Asymptotic comparisons

For functions on positive integers and $g(n) > 0$, write

$$f \in O(g) \iff \exists c > 0\ \exists N\ \forall n > N : |f(n)| \le c\,g(n).$$

The constants are independent of $n$. For positive $f, g$, the other standard comparisons are

$$f \in o(g) \iff f(n)/g(n) \to 0, \qquad f \in \Omega(g) \iff g \in O(f),$$
$$f \in \omega(g) \iff g \in o(f), \qquad f \in \Theta(g) \iff f \in O(g) \cap \Omega(g).$$

Also $f \sim g$ means $f(n)/g(n) \to 1$. For little-$o$, the quantified statement is $\forall \epsilon > 0\ \exists N\ \forall n > N : |f(n)| < \epsilon g(n)$; for little-$\omega$, every constant multiple of $g$ is eventually smaller than $f$.

If $f/g$ tends to a finite limit, then $f \in O(g)$. A positive finite limit establishes $\Theta(g)$, and an infinite positive limit establishes $\omega(g)$. These are sufficient tests for $O, \Omega, \Theta$, not necessary definitions: $2 + \sin n \in \Theta(1)$ despite having no limit. For example, $n \in o(n^2)$, every fixed-degree polynomial is in $o(2^n)$, $4^n \notin O(2^n)$, and $10n^3 + 20n^2 + 5 \in \Theta(n^3)$.

On the positive integers with $g$ everywhere positive, finitely many exceptional inputs can be absorbed into a larger constant, so $O(g)$ can equivalently be bounded at every input. This need not extend to continuous domains with unbounded behavior on a finite interval.

### 6.4 Using the notation correctly

The symbols describe sets of functions, so membership avoids treating an upper bound as an equality. Two functions in $O(g)$ need not equal one another. Little-$o$ implies $O$ and excludes $\Omega$; it is stronger than merely failing a lower bound. Nested notation also needs care: $2^{O(n)}$ denotes functions $2^{h(n)}$ for $h \in O(n)$, and includes $4^n$, whereas $O(2^n)$ does not. Upper bounds and lower bounds must be kept distinct when manipulating these expressions.

**Source.** Z. Abel, B. Chapman, and E. Demaine, *Mathematics for Computer Science*, Lecture 6 (MIT 6.1200J, Spring 2024). Adapted under CC BY-NC-SA 4.0.

# Lecture 7: Recurrences

## 7.1 Recursive descriptions

A recurrence specifies initial values and expresses later terms using earlier ones. For example, $a_0 = 1$, $a_n = a_{n-1} + 1$ has solution $a_n = n + 1$. The Fibonacci numbers $F_0 = 0$, $F_1 = 1$, $F_n = F_{n-1} + F_{n-2}$ have the closed form

$$F_n = \frac{1}{\sqrt{5}} \left[ \left( \frac{1+\sqrt{5}}{2} \right)^n - \left( \frac{1-\sqrt{5}}{2} \right)^n \right].$$

Initial conditions are part of the problem; a recurrence alone may have many solutions. A guessed formula can be verified by induction.

## 7.2 Towers of Hanoi

There are three posts and $n$ differently sized disks. Move one top disk at a time, never placing a larger disk on a smaller one. A state is a list of the posts containing disks $1, \dots, n$; moving disk $i$ requires all smaller disks to be on the third post.

**Algorithm 7.1: Move a Hanoi stack** $H(n, A, B)$

```
Input: n disks on post A; destination B; third post C
if n = 0 then
    return
H(n − 1, A, C)
Move disk n from A to B
H(n − 1, C, B)
```

The smaller-stack calls are correct by induction. The move count satisfies $T(0) = 0$, $T(n) = 2T(n-1)+1$, giving $T(n) = 2^n - 1$. The same number is necessary: before the largest disk first moves, all smaller disks must be moved to the remaining post; after its last move to the target, they must be moved onto it. Induction bounds both portions below by $T(n-1)$. For 64 disks the count exceeds $1.8 \cdot 10^{19}$ moves.

## 7.3 Sorting recurrences

Selection sort finds the largest of $n$ remaining elements and recurses on the other $n - 1$. Its comparison count is $T(n) = T(n-1) + n - 1$, $T(1) = 0$, hence $T(n) = n(n-1)/2$. Merge sort divides into two halves, sorts both recursively, and merges them by repeatedly taking the smaller first remaining element. Merging $n$ total elements uses at most $n - 1$ comparisons. For a power of two,

the worst-case count obeys

$$T(n) = 2T(n/2) + n - 1, \qquad T(1) = 0.$$

Repeated substitution gives

$$T(n) = \sum_{j=0}^{k-1} (n - 2^j) + 2^k T(n/2^k).$$

At $k = \log_2 n$, this becomes $n \log_2 n - n + 1$. For arbitrary sizes, use floor and ceiling halves; the asymptotic bound remains $\Theta(n \log n)$.

### 7.4 The master theorem

Consider $T(n) = aT(\lfloor n/b \rfloor) + f(n)$ above a fixed base threshold, with positive constant-size base costs, constants $a \geq 1$, $b > 1$, and nonnegative $f$. Put $d = \log_b a$.

**Theorem 7.1 [Master theorem].** *For a fixed $\epsilon > 0$:*

1. *If $f \in O(n^{d-\epsilon})$, then $T \in \Theta(n^d)$.*
2. *If $f \in \Theta(n^d)$, then $T \in \Theta(n^d \log n)$.*
3. *If $f \in \Omega(n^{d+\epsilon})$ and $af(\lfloor n/b \rfloor) \leq cf(n)$ eventually for some $c < 1$, then $T \in \Theta(f(n))$.*

*The analogous statement holds with ceiling subproblem sizes and the corresponding regularity condition.*

*Proof.* The recursion tree has $a^j$ nodes of size about $n/b^j$ at depth $j$, total work $a^j f(n/b^j)$ at that level, and depth $\Theta(\log n)$. In the first case, work grows geometrically toward the $\Theta(n^d)$ leaves. In the second, each level contributes $\Theta(n^d)$. In the third, the regularity condition bounds work on successive levels by a decreasing geometric series from the root. □

Merge sort uses the second case; binary search gives $\Theta(\log n)$ by the same case with $a = 1, b = 2, f = 1$. Karatsuba's recurrence $T(n) = 3T(\lceil n/2 \rceil) + \Theta(n)$ uses the first case and gives $\Theta(n^{\log_2 3})$.

The theorem requires its stated form. It does not apply to Hanoi's $n - 1$ subproblem. Its basic three cases also leave gaps, such as $T(n) = 2T(n/2) + n \log n$: a logarithmic excess is not the polynomial gap required in case three. A recursion tree or substitution is needed there.

**Source.** Z. Abel, B. Chapman, and E. Demaine, *Mathematics for Computer Science*, Lecture 7 (MIT 6.1200J, Spring 2024). Adapted under CC BY-NC-SA 4.0.

# Lecture 8: Divisibility

## 8.1 Divisors and integer linear combinations

For integers $a, b$, write $a \mid b$ when $b = ak$ for some integer $k$. This definition includes $0 \mid 0$; it does not require division by $a$. Divisibility is transitive, and if $a \mid b$ and $a \mid c$, then $a \mid sb + tc$ for every $s, t \in \mathbb{Z}$. An expression $sb + tc$ with integer coefficients is an integer linear combination of $b, c$.

Two jugs of capacities $a, b$, initially empty, can be filled, emptied, or poured into one another until a source empties or a destination fills. A state is the pair of amounts $(x, y)$. Every reachable amount is an integer linear combination of $a, b$: the property holds at $(0, 0)$, and a transition replaces coordinates by members of $\{0, x, y, a, b, x + y, x + y - a, x + y - b\}$, preserving it. Thus capacities $6, 9$ cannot produce $5$, since every reachable amount is divisible by $3$. Capacities $3, 5$ can produce $4$, for example

$$(0, 0) \to (0, 5) \to (3, 2) \to (0, 2) \to (2, 0) \to (2, 5) \to (3, 4).$$

## 8.2 Greatest common divisors and remainders

The greatest common divisor $\gcd(a, b)$ is the nonnegative common divisor divisible by every common divisor of $a, b$. This definition gives $\gcd(a, 0) = |a|$ and $\gcd(0, 0) = 0$.

For an integer $n$ and positive $d$, the division theorem gives unique integers $q, r$ with $n = qd + r$ and $0 \le r < d$. Write $q = \lfloor n/d \rfloor$ and $r = n \text{ rem } d$. The pairs $(a, b)$ and $(a, b - qa)$ have exactly the same common divisors, because a divisor of two integers divides every integer linear combination of them. In particular,

$$\gcd(a, b) = \gcd(b, a \text{ rem } b) \quad (b > 0).$$

## 8.3 Euclid's algorithm

Start with $x = |a|$, $y = |b|$. While $y > 0$, replace $(x, y)$ by $(y, x \text{ rem } y)$; return $x$ when $y = 0$. The gcd remains invariant, and $\gcd(x, 0) = x$ proves correctness at termination. After a possible initial swap, $x \ge y$ and the nonnegative potential $x + y$ strictly decreases.

The number of divisions is logarithmic in the input magnitudes. Indeed, for $x \ge y > 0$, the remainder is less than $x/2$: if $y \le x/2$ it is less than $y$, and otherwise it equals $x - y < x/2$. Thus the total binary length of the two coordinates decreases by at least one per division. This counts division operations, rather than claiming that arbitrary-size integer division has unit bit cost.

For example, the gcd of 1001 and 777 is 7:

$$(1001, 777) \to (777, 224) \to (224, 105) \to (105, 14) \to (14, 7) \to (7, 0),$$

## 8.4 The extended algorithm and Bézout's identity

Maintain $x = as + bt$ and $y = au + bv$, starting with coefficient pairs $(s, t) = (1, 0)$ and $(u, v) = (0, 1)$ for nonnegative inputs. If $x = qy + r$, update

$$(x, y, s, t, u, v) \leftarrow (y, r, u, v, s - qu, t - qv).$$

The coefficient invariants follow from $r = a(s - qu) + b(t - qv)$. At termination,

$$\gcd(a, b) = as + bt.$$

This is Bézout's identity; negative inputs can be handled by adjusting coefficient signs. In the example, $7 = -52(1001) + 67(777)$.

Every integer linear combination of $a, b$ is a multiple of their gcd. Conversely, multiplying Bézout's identity shows that every multiple of the gcd is an integer linear combination. If the inputs are not both zero, their gcd is therefore the smallest positive such combination.

**Source.** Z. Abel, B. Chapman, and E. Demaine, *Mathematics for Computer Science*, Lecture 8 (MIT 6.1200J, Spring 2024). Adapted under CC BY-NC-SA 4.0.

# Lecture 9: Modular Arithmetic

## 9.1 Congruence and residues

For a positive modulus $n$, define $a \equiv_n b$ when $n \mid a - b$. Writing $a = qn + r$ with $0 \le r < n$ gives $a \equiv_n r$. Two integers are congruent exactly when their nonnegative remainders agree, so $0, \ldots, n-1$ represent the $n$ residue classes. Congruence is a relation; the remainder operation returns one specific representative. For example, $12 \equiv_5 17$ although $12 \neq 17$.

Clock times, weekdays, last digits, and parity use moduli 12 or 24, 7, 10, and 2. We always take nonnegative remainders: for example $-43 \text{ rem } 10 = 7$.

## 9.2 Arithmetic and exponentiation

Congruences may be added, subtracted, and multiplied. For instance, if $a \equiv_n b$, then $ac - bc = c(a-b)$ is divisible by $n$, so $ac \equiv_n bc$. Repeated multiplication also gives $a^k \equiv_n b^k$ for integer $k \ge 0$. Thus intermediate values can be replaced by residues to keep calculations small. Exponents cannot generally be reduced modulo $n$: $1 \equiv_5 6$ but $2^1 \not\equiv_5 2^6$.

For example, the last two digits of $11335^{11111}(6 + 7799^{5000})$ are found modulo 100. The second base is $-1$ modulo 100, so its even power is 1. Powers of 35 alternate between 25 and 75 after the first power, giving $35^{11111} \equiv_{100} 75$. The product is $75 \cdot 7 \equiv_{100} 25$.

## 9.3 Inverses and cancellation

A multiplicative inverse of $a$ modulo $n$ is an integer $b$ satisfying $ab \equiv_n 1$.

**Theorem 9.1.** *An inverse exists exactly when* $\gcd(a, n) = 1$.

*Proof.* The equation $ab \equiv_n 1$ is equivalent to $ab + nq = 1$ for some integer $q$. Such a linear combination exists exactly when the gcd divides 1, by Bézout's identity. □

The extended Euclidean algorithm computes an inverse. Cancellation is valid for invertible factors, but not in general: $3x \equiv_6 3$ allows $x = 5$ as well as $x = 1$. Modulo 30, $7^{-1} \equiv_{30} 13$; thus $7x \equiv_{30} 12$ gives $x \equiv_{30} 13 \cdot 12 \equiv_{30} 6$. For prime $p$, every nonzero residue has an inverse.

## 9.4 Fermat's little theorem

**Theorem 9.2.** *If $p$ is prime and $p \nmid a$, then $a^{p-1} \equiv_p 1$.*

*Proof.* Multiplication by $a$ permutes the nonzero residues: cancellation shows that $ai \equiv_p aj$ forces $i = j$ for $1 \le i, j < p$. Taking products gives $a^{p-1}(p-1)! \equiv_p (p-1)!$. The factorial is invertible modulo $p$, so it can be canceled. □

For a base coprime to prime $p$, exponents may therefore be reduced modulo $p-1$, rather than modulo $p$.

## 9.5 Digit tests and checksums

If $N = \sum_i d_i 10^i$, then $10^i \equiv_9 1$ gives $N \equiv_9 \sum_i d_i$. This proves the usual divisibility-by-nine test and also preserves the entire residue modulo 9.

The older ISBN-10 checksum satisfies $\sum_{i=1}^{10} i a_i \equiv_{11} 0$. Because 10 is invertible modulo 11, the first nine digits determine the last check symbol; value 10 is written as X. Changing one symbol changes the weighted sum by a nonzero product modulo 11. Swapping unequal adjacent symbols changes it by their nonzero difference, so both errors are detected.

Redundancy can also recover missing data: storing bits $b_1, b_2$ and $b_3 = b_1 \oplus b_2$ permits recovery of a missing $b_2$ as $b_1 \oplus b_3$. Here $\oplus$ is addition modulo 2.

**Source.** Z. Abel, B. Chapman, and E. Demaine, *Mathematics for Computer Science*, Lecture 9 (MIT 6.1200J, Spring 2024). Adapted under CC BY-NC-SA 4.0.

# Lecture 10: Cryptography

## 10.1 Encryption and secret keys

An encryption scheme transforms a message into a ciphertext that the intended recipient can decrypt. The participants conventionally are Alice, Bob, and an eavesdropper Eve. Security should depend on a secret key rather than on hiding the algorithm.

A Caesar shift sends each letter to its value plus $k$ modulo 26; decryption subtracts $k$. Only 26 keys exist, so brute force is easy. A substitution cipher has 26! keys but retains letter-frequency patterns. A large key space alone therefore does not establish security. Known plaintext can expose structural weaknesses even in a changing substitution system, as illustrated by attacks on Enigma.

## 10.2 One-time pads

Represent a message by $m \in \{0, \ldots, n-1\}$. Choose a uniformly random independent key $k$ from the same set, encrypt as $c = (m + k) \text{ rem } n$, and decrypt by subtracting $k$. For every fixed $m$ and possible $c$, exactly one key produces $c$, so the ciphertext distribution is uniform and reveals no information about the message.

The key must be used once. Reuse gives $c_1 - c_2 \equiv_n m_1 - m_2$, revealing a relation between messages; one known message also reveals the key. Securely sharing a sufficiently long fresh key is the remaining difficulty.

## 10.3 Diffie–Hellman key agreement

Choose a suitable prime modulus $p$ and public group element $g$. Alice privately chooses $a$ and publishes $A = g^a \bmod p$; Bob privately chooses $b$ and publishes $B = g^b \bmod p$. They compute

$$B^a \equiv_p (g^b)^a \equiv_p g^{ab} \equiv_p (g^a)^b \equiv_p A^b,$$

so their shared value agrees. Eve sees $p, g, A, B$. Recovering an exponent from its group power is the discrete logarithm problem. Shared-value security requires an appropriate computational Diffie–Hellman assumption and suitable parameters, rather than just hardness of recovering exponents. Bare key agreement does not authenticate the senders.

Repeated squaring evaluates $g^a \bmod p$ in $O(\log a)$ modular multiplications. Read the exponent's binary digits, squaring the accumulated power and multiplying by $g$ when the next digit is one. Reduce modulo $p$ after each operation to avoid constructing $g^a$.

## 10.4 RSA

Choose distinct large primes $p, q$ and set $n = pq$. Choose $e$ coprime to $\phi = (p-1)(q-1)$ and use the extended Euclidean algorithm to find $d$ satisfying $ed \equiv_\phi 1$. Publish $(n, e)$ and keep $d, p, q$ secret. The textbook transforms are

$$c = m^e \bmod n, \qquad m = c^d \bmod n, \qquad 0 \le m < n.$$

To verify correctness, write $ed = 1 + t(p-1)(q-1)$. Modulo $p$, either $m \equiv_p 0$, so $m^{ed} \equiv_p m$, or Fermat's little theorem gives

$$m^{ed} = m\left(m^{p-1}\right)^{t(q-1)} \equiv_p m.$$

The same argument holds modulo $q$. The Chinese remainder theorem then gives $m^{ed} \equiv_n m$, including messages sharing a factor with $n$.

Factoring $n$ yields $p, q, d$, but its assumed hardness does not prove RSA inversion hard. The RSA assumption concerns extracting $e$th roots modulo $n$ directly. Neither assumption is a proven computational lower bound. The deterministic textbook transform also needs appropriate encoding and protocol precautions for secure encryption.

Large primes can be found by sampling candidates and applying a primality test. The prime number theorem states that the number of primes up to $x$ is asymptotic to $x/\ln x$, motivating a prime density about $1/\ln x$ near large $x$. Randomized primality tests such as Miller–Rabin make this search practical.

## 10.5 Chinese remainder theorem

**Theorem 10.1 [Chinese remainder theorem].** *If positive integers $p, q$ are coprime, then for every $a, b$ there is a unique residue $x$ modulo $pq$ with $x \equiv_p a$ and $x \equiv_q b$.*

*Proof.* Let $u = q^{-1} \bmod p$ and $v = p^{-1} \bmod q$. Then $x = auq + bvp$ has the required two residues. If $x'$ is another solution, $y = x - x'$ is divisible by both $p, q$. Choose integers $s, t$ with $ps + qt = 1$. In $y = psy + qty$, both terms are divisible by $pq$, so $x \equiv_{pq} x'$. □

For example, $x \equiv_5 4$ and $x \equiv_{11} 7$ determine $x \equiv_{55} 29$. The theorem combines two smaller modular calculations into one calculation modulo their product.

**Source.** Z. Abel, B. Chapman, and E. Demaine, *Mathematics for Computer Science*, Lecture 10 (MIT 6.1200J, Spring 2024). Adapted under CC BY-NC-SA 4.0.

# Lecture 11: Graphs and Coloring

## 11.1 Degrees and bipartitions

A finite simple graph is $G = (V, E)$, where $V$ is a nonempty vertex set and $E$ consists of two-element subsets of $V$. Thus edges are undirected, with no loops or repeated edges. Vertices $u, v$ are adjacent when $\{u, v\} \in E$; that edge is incident to both endpoints. The degree $\deg(v)$ counts incident edges.

**Theorem 11.1 [Handshake lemma].** *For every finite undirected graph,* $\sum_{v \in V} \deg(v) = 2|E|$.

*Proof.* Count pairs $(v, e)$ in which $v$ is an endpoint of $e$. Counting by vertices gives the sum of degrees; counting by edges gives two per edge. □

Consequently the number of odd-degree vertices is even, and a simple $n$-vertex graph has at most $n(n-1)/2$ edges. The complete graph $K_n$ attains this bound, since every vertex has degree $n - 1$.

A graph is bipartite if $V$ can be partitioned into disjoint sets $L, R$ with every edge joining the two sets. Counting each edge by its left and right endpoints gives

$$\sum_{v \in L} \deg(v) = |E| = \sum_{v \in R} \deg(v).$$

For nonempty parts and at least one edge, the ratio of average degrees on $L$ and $R$ is $|R|/|L|$. This conclusion depends only on the part sizes, not on the particular edge pattern.

## 11.2 Coloring as a conflict model

A proper $k$-coloring assigns one of $k$ colors to each vertex so adjacent vertices receive different colors. The chromatic number $\chi(G)$ is the smallest possible $k$. To prove $\chi(G) = k$, one needs both a coloring using $k$ colors and an obstruction to fewer.

For exam scheduling, use one vertex per course and an edge when two courses share a student. Colors represent time slots. A triangle forces three slots; producing a three-coloring proves three suffice. The same model describes register allocation, radio-frequency conflicts, and map coloring. A proposed coloring can be checked by examining all edges, although deciding whether an arbitrary graph is three-colorable is NP-complete.

## 11.3 Greedy coloring and its limits

Order the vertices, then give each vertex the smallest positive color not already assigned to a colored neighbor. Different orders can produce different numbers of colors.

**Theorem 11.2.** *If every vertex has degree at most $\Delta$, greedy coloring in any order uses at most $\Delta + 1$ colors. In particular, $\chi(G) \le \Delta + 1$.*

*Proof.* When a vertex is processed, at most $\Delta$ of its neighbors have already been colored. They exclude at most $\Delta$ colors, leaving one of the first $\Delta + 1$ available. Equivalently, induct on the number of vertices: remove the final vertex of the specified order, color the remaining graph, and restore it with an available color. □

For graph induction, start with an arbitrary graph of size $n + 1$ and reduce it to size $n$. Building larger graphs from just one chosen $n$-vertex graph need not cover every case.

The bound is tight for $K_{\Delta+1}$. It can also be loose: the star $K_{1,k}$ has maximum degree $k$ but requires only two colors, which greedy coloring finds in every order. More seriously, greedy coloring can perform poorly even on bipartite graphs.

Take left vertices $\ell_1, \dots, \ell_n$, right vertices $r_1, \dots, r_n$, and edges $\ell_i r_j$ exactly when $i \neq j$. For $n \ge 2$ this CROWN GRAPH has chromatic number two. In the order $\ell_1, r_1, \ell_2, r_2, \dots, \ell_n, r_n$, however, the $i$th pair sees every earlier color on opposite-side neighbors, so both receive new color $i$. Thus greedy uses $n$ colors.

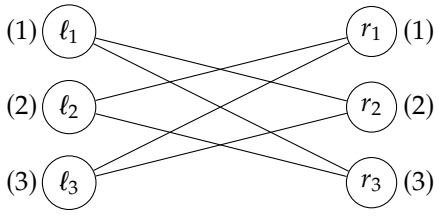



**Figure 2:** A crown graph with $n = 3$. Labels in parentheses are greedy colors in the alternating order. Coloring by side needs only two.

**Source.** Z. Abel, B. Chapman, and E. Demaine, *Mathematics for Computer Science*, Lecture 11 (MIT 6.1200J, Spring 2024). Adapted under CC BY-NC-SA 4.0.

# Lecture 12: Matching

## 12.1 Size, maximality, and weights

A matching is a set of edges with no shared endpoints. It is maximal if no edge can be added while retaining this property, and maximum if no matching has larger size. These differ: in a four-vertex path, the middle edge alone is maximal, whereas the two outer edges form a larger matching. A perfect matching covers every vertex and therefore has $|V|/2$ edges. Two degree-one vertices with the same neighbor prevent perfection, because at least one must remain unmatched.

In a bipartite graph, a perfect matching requires equal part sizes; equal sizes alone do not guarantee its existence. For a weighted graph, $w(M) = \sum_{e \in M} w(e)$. Minimum-weight perfect matching minimizes this sum among perfect matchings. Missing edges can be modeled as having infinite weight. For two left and two right vertices with edge-weight matrix

$$\begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix},$$

the diagonal perfect matching has weight 5, whereas the other has weight 6. Pairing problems model assignments such as tasks to servers or aircraft to gates.

## 12.2 Stable matching

Consider $N$ applicants and $N$ evaluators. Each participant strictly ranks every member of the opposite side, with no ties. Preferences need not be symmetric. An unmatched pair $(a, e)$ is a blocking pair if both prefer one another to their assigned partners. A perfect matching is stable when it has no blocking pair. Stability is a condition on individual preferences, rather than on a total edge weight.

The bipartite assumption matters. Suppose three participants have cyclic preferences $a : b > c > d$, $b : c > a > d$, $c : a > b > d$, while $d$'s order is arbitrary. In any perfect pairing, one of $a, b, c$ is paired with $d$. If it is $a$, the other pair is $(b, c)$, but $(a, c)$ blocks. The two other cases follow by cyclic relabeling. A stable perfect matching therefore need not exist in the nonbipartite setting.

## 12.3 Gale–Shapley deferred acceptance

Initially all applicants are free. A free applicant proposes to its favorite untried evaluator, which holds the better of its current applicant and the new proposer. The rejected applicant becomes free. Held pairs are tentative until termination.

**Algorithm 12.1: Applicant-proposing deferred acceptance**

```
while some applicant a is free and has an untried evaluator do
    Let e be a's highest-ranked untried evaluator
    Mark the proposal (a, e) as tried
    if e holds no applicant then
        e holds a
    else
        e holds its preferred applicant among a and its current hold
        The other applicant is free
return the held pairs
```

**Theorem 12.1.** *With complete strict lists and equal part sizes, deferred acceptance terminates with a stable perfect matching after at most $N^2$ proposals.*

*Proof.* No applicant proposes to the same evaluator twice, so there are at most $N^2$ proposals. Once an evaluator holds someone, it never becomes empty, and its held applicant can only improve in its preference order. An applicant's successive proposals move downward through its list.

If an applicant were unmatched at termination, it would have exhausted all evaluators. Each evaluator would then hold someone, giving $N$ matched applicants in addition to the unmatched one, a contradiction. For stability, if $a$ preferred $e$ to its final partner, $a$ must have proposed to $e$ and been rejected. Since held partners only improve, $e$ prefers its final partner to $a$. Thus $(a, e)$ cannot block. □

The equivalent simultaneous-day version has each applicant return to its favorite evaluator that has not rejected it. Every nonterminal day adds a rejection, so the number of unrejected list entries is a strictly decreasing nonnegative potential. This gives the stated upper bound $N^2 + 1$ days; days and individual proposals are different cost measures.

## 12.4 Which stable matching is selected?

A FEASIBLE PARTNER of a participant is a partner it receives in some stable matching. Applicant-proposing Gale–Shapley gives every applicant its best feasible partner and every evaluator its worst feasible partner. Thus stability does not mean that the two sides receive symmetric outcomes. Swapping the proposing side swaps these extremal guarantees.

**Source.** Z. Abel, B. Chapman, and E. Demaine, *Mathematics for Computer Science*, Lecture 12 (MIT 6.1200J, Spring 2024). Adapted under CC BY-NC-SA 4.0.

# Lecture 13: Connectivity and Trees

## 13.1 Walks, paths, and components

A walk $(v_0, \ldots, v_k)$ follows an edge between each successive pair; its length is $k$, although the sequence has $k+1$ vertices. A trail repeats no edge, and a path repeats no vertex. Vertices are connected when a walk joins them. The length-zero walk $(v)$ connects every vertex to itself. A graph is connected if every pair is connected, a weaker requirement than every pair being adjacent.

If a walk joins $a$ to $b$, a path does too: choose a shortest walk. Any repeated vertex would enclose a segment that could be deleted, contradicting minimality. Connectivity is reflexive, symmetric, and transitive, so its equivalence classes partition the vertices. The subgraph induced by each class is a connected component. Every edge lies within one component.

## 13.2 Euler trails and tours

A walk is closed when its endpoints agree. A tour is a closed trail. In a simple undirected graph, a cycle is a closed walk of length at least three with no repetition other than the initial/final vertex. An Euler tour uses every edge exactly once and visits every vertex; an open Euler trail has the same coverage but different endpoints. Such a trail may revisit vertices, despite sometimes being called an Euler path.

**Theorem 13.1.** *A finite connected graph has an Euler tour exactly when all degrees are even. It has an open Euler trail exactly when two degrees are odd.*

*Proof.* In a tour, each arrival at a vertex is paired with a departure, giving even degree. For sufficiency, choose a longest trail $W$. If its endpoints differed, the final vertex would have used an odd number of incident edges and would have an unused edge, extending $W$. Thus $W$ is closed. If any edge remained unused, connectivity would give a vertex on $W$ incident to an unused edge. Starting the closed trail there and then taking this edge again extends it, a contradiction.
For an open trail, only its endpoints have unpaired arrivals or departures, so exactly those two degrees are odd. Conversely, add an auxiliary vertex $z$ and edges $uz, zv$ between the two odd-degree vertices $u, v$. The new connected graph has all degrees even. Remove the segment $u, z, v$ from an Euler tour to obtain an open Euler trail. □

The bridges of Königsberg give a multigraph with four odd-degree vertices, so neither kind of Euler traversal exists. The parity argument applies to multigraphs as well as simple graphs.

### 13.3 Trees and leaf removal

A TREE is a connected acyclic graph. A FOREST is any acyclic graph, whose connected components are trees.

**Lemma 13.2.** *Every finite tree with at least two vertices has at least two leaves, where a leaf has degree one.*

*Proof.* Take a longest path. An endpoint's additional neighbor cannot already be on the path, since that would create a cycle, and cannot be outside it, since that would extend the path. Both endpoints are leaves. □

Removing a leaf and its incident edge preserves acyclicity and connectivity: a path between other vertices could not have used a degree-one vertex internally. Induction, starting with the one-vertex tree and zero edges, therefore proves that an $n$-vertex tree has $n - 1$ edges. Removing an arbitrary vertex need not preserve connectivity.

For a finite simple graph on $n \geq 1$ vertices, the following characterizations of trees are equivalent:

- connected and acyclic;
- connected with exactly $n - 1$ edges;
- acyclic with exactly $n - 1$ edges;
- every pair of vertices has a unique path between them;
- minimally connected: deleting any edge disconnects it;
- maximally acyclic: adding any missing edge creates a cycle.

Two distinct paths between a pair would create a cycle. Conversely, if an edge lies on a cycle, deleting it preserves connectivity. These observations explain why trees have no redundant connections.

**Source.** Z. Abel, B. Chapman, and E. Demaine, *Mathematics for Computer Science*, Lecture 13 (MIT 6.1200J, Spring 2024). Adapted under CC BY-NC-SA 4.0.

# Lecture 14: Digraphs and DAGs

## 14.1 Direction and reachability

A directed graph has $E \subseteq V \times V$; an ordered edge $(u, v)$ points from $u$ to $v$. Loops and opposite edges $(u, v), (v, u)$ are allowed, but repeated copies of the same ordered edge are not. State-machine transitions and prerequisite constraints are directed edges. A two-way connection uses two directed edges.

The indegree $\deg^-(v)$ counts incoming edges and outdegree $\deg^+(v)$ counts outgoing edges. A loop contributes one to each. Counting endpoints gives

$$\sum_v \deg^-(v) = |E| = \sum_v \deg^+(v).$$

Directed walks, trails, and paths must follow every arrow. A shortest walk again has no repeated vertex, so reachability implies a path. A loop is a directed cycle of length one; opposite edges give a cycle of length two. Reachability need not be symmetric.

Vertices $u, v$ are strongly connected if each can reach the other. The strongly connected components (SCCs) partition the vertices, although edges can run between different SCCs. Collapse each SCC to a vertex and keep edges between distinct components: this is the condensation graph. It has no directed cycle, because components on such a cycle would all reach one another and would constitute a single SCC.

A directed graph has an Euler tour visiting all vertices exactly when it is strongly connected and $\deg^-(v) = \deg^+(v)$ for every vertex. Necessity follows by balancing arrivals and departures. For sufficiency, the longest-trail argument works as before: balance forces closure, and strong connectivity prevents unused edges from being inaccessible.

## 14.2 Topological orders

A directed acyclic graph (DAG) has no directed cycle. A source has indegree zero and a sink has outdegree zero. A finite nonempty DAG has both: the endpoints of a longest directed path must be a source and sink, since otherwise an extra edge either extends the path or creates a cycle.

A topological order lists all vertices with every edge pointing forward. Every finite DAG has one. Repeatedly remove a source and append it to the list; deleting vertices cannot create a cycle, so a source remains available until the graph is empty. Equivalently, induct on the number of vertices and prepend a source to an order of the remaining graph.

An edge is REDUNDANT if another directed path joins its endpoints. Removing redundant edges from a DAG leaves its reachability relation unchanged. Its remaining covering edges give the Hasse diagram of the corresponding precedence order. For example, a direct shirt-to-scarf edge is redundant when shirt → jacket → scarf already enforces the constraint.

### 14.3 Parallel scheduling, chains, and antichains

Assume each task takes one stage and arbitrarily many helpers are available. Two vertices are comparable if one reaches the other. A CHAIN is a set with every pair comparable; an ANTICHAIN has every distinct pair incomparable. Tasks performed together must form an antichain. A largest chain, called a critical path here, has size $c$ in *vertices*; the corresponding longest path has $c - 1$ edges.

**Theorem 14.1.** *The minimum number of parallel stages is the largest chain size $c$.*

*Proof.* The $c$ tasks in a chain must occupy distinct stages. For the reverse bound, let $d(v)$ be the maximum number of edges in a path ending at $v$. Along any nonempty directed path from $u$ to $v$, $d(v) > d(u)$. Process all vertices of depth $i$ at stage $i + 1$. Every prerequisite has smaller depth, and depths range from 0 to $c - 1$, giving $c$ valid stages. □

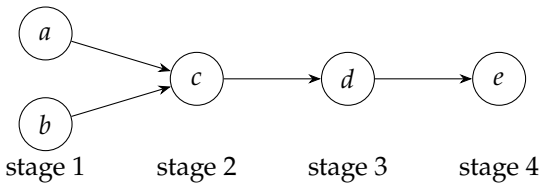



**Figure 3:** Depth layers are parallel stages. The chain $a, c, d, e$ forces four stages, even with unlimited helpers.

Scheduling every available task immediately minimizes stages with unlimited helpers, but need not minimize the helpers needed for that same number of stages. The largest antichain bounds simultaneous work.

If the largest antichain has size $w$, the depth partition has $c$ antichains, so $n \le cw$. Hence, for every $t > 0$, either $c > t$ or $w \ge n/t$. In particular, there is a chain or antichain of size at least $\sqrt{n}$. This chain/antichain size bound follows directly from the depth partition.

**Source.** Z. Abel, B. Chapman, and E. Demaine, *Mathematics for Computer Science*, Lecture 14 (MIT 6.1200J, Spring 2024). Adapted under CC BY-NC-SA 4.0.

# Lecture 15: Relations and Counting

## 15.1 Relations, functions, and set sizes

A relation from domain $A$ to codomain $B$ is a subset $R \subseteq A \times B$. Write $aRb$ or $R(a, b)$ for $(a, b) \in R$. Its pairs can be drawn as arrows from $A$ to $B$. Use the following conventions, including *partial* functions:

| Property | Arrow condition |
|---|---|
| Function | At most one outgoing arrow per $a \in A$ |
| Total | At least one outgoing arrow per $a \in A$ |
| Injective | At most one incoming arrow per $b \in B$ |
| Surjective | At least one incoming arrow per $b \in B$ |

A total function gives exactly one output for every input. A bijection is a total, injective, surjective function. Properties depend on the declared domain and codomain: $x \mapsto 1/x^2$ is partial on $\mathbb{R}$, but total on $\mathbb{R} \setminus \{0\}$.

For finite sets, a total injective relation implies $|A| \leq |R| \leq |B|$. A surjective function gives $|B| \leq |R| \leq |A|$. A bijection therefore proves equal cardinality. These are counting arguments about arrows, so they apply even without an explicit formula for a function.

## 15.2 Equivalence and order

A relation on one set is also a directed graph. It is reflexive if $aRa$ always holds, symmetric if $aRb$ implies $bRa$, and transitive if $aRb$ and $bRc$ imply $aRc$.

An equivalence relation has all three properties. Its classes $[a] = \{b : aRb\}$ partition the domain: if two classes share an element, symmetry and transitivity make the classes equal. Examples include congruence modulo an integer and strong connectivity.

Antisymmetry instead says ($aRb$ and $bRa$) $\Rightarrow a = b$. A weak partial order is reflexive, antisymmetric, and transitive. Examples are subset inclusion, divisibility on positive integers, and reachability in a DAG, including length-zero walks. Antisymmetry is different from requiring that reverse pairs never occur, since self-pairs are allowed. A total order is a partial order in which every pair is comparable.

### 15.3 Basic counting rules

For finite sets, the product rule gives $|A_1 \times \cdots \times A_k| = \prod_i |A_i|$. For pairwise disjoint sets, the sum rule gives $|\bigcup_i A_i| = \sum_i |A_i|$. A bijection transfers the count from one set to another. For example, subsets of $[n] = \{1, \ldots, n\}$ correspond to $n$-bit membership strings, so there are $2^n$ subsets.

As an application combining sum and product, strings of length six through eight with an uppercase initial letter and subsequent letters or digits number

$$26\left(62^5 + 62^6 + 62^7\right).$$

Lengths are disjoint cases; the character positions are successive choices within each case.

The GENERALIZED PRODUCT RULE permits dependent choice *sets*, provided their *sizes* stay fixed: if every legal prefix of length $i - 1$ has exactly $n_i$ possible next entries, the number of complete sequences is $\prod_i n_i$. Thus $n$ distinct objects have $n!$ permutations. Eight-digit strings with no repeated digit number $10 \cdot 9 \cdots 3$, even though the available next digits depend on the preceding choices.

**Source.** Z. Abel, B. Chapman, and E. Demaine, *Mathematics for Computer Science*, Lecture 15 (MIT 6.1200J, Spring 2024). Adapted under CC BY-NC-SA 4.0.

# Lecture 16: Counting Techniques

## 16.1 Division and combinations

A counting recipe should produce every desired object and identify how often each is produced. If a total map $f : A \to B$ has exactly $k$ preimages for every $b \in B$, its fibers partition $A$ into sets of size $k$, giving the division rule $|B| = |A|/k$.

For $n \geq 1$ distinct people at a round table, identify rotations but not reflections. Each cyclic order comes from $n$ linear orders, giving $n!/n = (n-1)!$ cyclic orders. Similarly, an ordered selection of $r$ distinct elements has $n!/(n-r)!$ possibilities. Every $r$-element subset appears in $r!$ orders, so

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}, \qquad 0 \leq r \leq n.$$

There are $\binom{10}{3} = 120$ strictly increasing three-digit strings, allowing an initial zero, since choosing the three digits uniquely determines their increasing order. Directly multiplying the numbers of possible digits at each position fails here: those numbers depend on the preceding digits. Likewise, $\binom{100}{50}$ bit strings of length 100 have exactly 50 ones.

## 16.2 Poker-hand recipes

A five-card hand is an unordered subset of a deck with 13 ranks and four suits, so there are $\binom{52}{5}$ hands. The following recipes distinguish bijections from uniform and nonuniform overcounting.

**Four of a kind.** Choose its rank in 13 ways and its one extra card in 48 ways. Each desired hand has exactly one such description, giving $13 \cdot 48 = 624$ hands.

**All four suits.** Exactly one suit occurs twice. Choose that suit, its two ranks, and one rank for each other suit:

$$4\binom{13}{2}13^3.$$

Alternatively choose one card of each suit and then an extra card, giving $13^4 \cdot 48$ descriptions. Each resulting hand has exactly two descriptions, according to which card in its repeated suit was chosen as the extra. Dividing by two gives the same count.

**At least one repeated rank.** Choosing a distinguished pair and then three remaining cards gives $13\binom{4}{2}\binom{50}{3}$ descriptions, but multiplicity varies: two pairs give two descriptions, a triple gives three, a full house gives four, and four of a kind gives six. No single division corrects this overcount.

Instead count the complement. A hand with no repeated rank chooses five distinct ranks and then one suit for each:

$$N_{\text{no pair}} = \binom{13}{5}4^5, \qquad N_{\text{repeated rank}} = \binom{52}{5} - \binom{13}{5}4^5.$$

The equivalent ordered recipe for the complement is $52 \cdot 48 \cdot 44 \cdot 40 \cdot 36/5!$. The essential checks are coverage and multiplicity: producing only valid objects is not enough to establish a correct count.

**Source.** Z. Abel, B. Chapman, and E. Demaine, *Mathematics for Computer Science*, Lecture 16 (MIT 6.1200J, Spring 2024). Adapted under CC BY-NC-SA 4.0.

# Lecture 17: More Counting Techniques

## 17.1 Inclusion–exclusion

For overlapping finite sets, adding cardinalities counts intersections too often. In particular,

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

For three sets subtract all pairwise intersections and add back the *triple intersection*. In general, writing $[n] = \{1, \dots, n\}$,

**Theorem 17.1 [Inclusion–exclusion].**

$$\left| \bigcup_{i=1}^{n} A_i \right| = \sum_{\varnothing \neq I \subseteq [n]} (-1)^{|I|+1} \left| \bigcap_{i \in I} A_i \right|.$$

*Proof.* For an element belonging to $r \geq 1$ sets, its net coefficient on the right is $\sum_{j=1}^{r} (-1)^{j+1} \binom{r}{j} = 1$. One can see the cancellation without algebra by toggling one fixed containing set, pairing even and odd subsets of its containing indices. Elements in no set contribute zero. □

There are $4 + 13 - 1 = 16$ cards that are queens or hearts. For distinct primes $p, q, r$ and $n = pqr$, apply the formula to multiples of these primes in $\{1, \dots, n\}$. The number coprime to $n$ is

$$n - \frac{n}{p} - \frac{n}{q} - \frac{n}{r} + \frac{n}{pq} + \frac{n}{pr} + \frac{n}{qr} - \frac{n}{pqr} = (p-1)(q-1)(r-1).$$

## 17.2 Pigeonhole arguments

If a total function maps a finite set $A$ into a smaller set $B$, two distinct inputs have the same output. More generally, $|A| > k|B|$ forces some output to have at least $k + 1$ preimages: otherwise counting all fibers gives $|A| \leq k|B|$. The conclusion proves a collision exists without necessarily locating it.

Picking $n + 1$ socks from $n$ color pairs guarantees two of one color, while $n$ picks can all differ. No lossless compressor can strictly shorten every $n$-bit string: there are $2^n$ inputs but only $1 + 2 + \cdots + 2^{n-1} = 2^n - 1$ shorter outputs, including the empty string.

For a less direct application, place rooks in 33 cells of an $8 \times 8$ board. Label cell $(i, j)$ by $(i + j) \bmod 8$. Each label appears once per row and column. Among eight labels, some labels at least five rooks, which are therefore in distinct rows and columns and do not attack one another.

### 17.3 Combinatorial proofs and expansions

Counting all subsets by size gives $\sum_{k=0}^{n} \binom{n}{k} = 2^n$. Counting $k$-subsets according to whether they contain the final element gives Pascal's identity

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

Here a coefficient is zero when its lower index is outside the range $0, \ldots, n-1$. This explains the adjacent-entry recurrence in Pascal's triangle.

Expanding a product of $n$ copies of $x + y$, the term $x^k y^{n-k}$ arises by choosing which $k$ factors contribute $x$. Thus

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.$$

The same argument with $m$ choices per factor gives

$$(x_1 + \cdots + x_m)^n = \sum_{\substack{k_1+\cdots+k_m=n \\ k_i \geq 0}} \frac{n!}{k_1! \cdots k_m!} \prod_{i=1}^{m} x_i^{k_i}.$$

The multinomial coefficient counts assignments of positions to the $m$ categories. Empty products equal 1, empty sums equal 0, and zero powers in these expansions use $0^0 = 1$.

For another double count, select $n$ elements from two disjoint $n$-element sets. Choosing $k$ from the first forces $n - k$ from the second, so

$$\sum_{k=0}^{n} \binom{n}{k}^2 = \binom{2n}{n}.$$

**Source.** Z. Abel, B. Chapman, and E. Demaine, *Mathematics for Computer Science*, Lecture 17 (MIT 6.1200J, Spring 2024). Adapted under CC BY-NC-SA 4.0.

# Lecture 18: Introduction to Probability

## 18.1 Discrete spaces and the tree method

A discrete probability space consists of a nonempty countable sample space $S$ and masses $p(\omega) \geq 0$ summing to one. An outcome is an element $\omega \in S$; an event is a subset $A \subseteq S$, with $\mathbb{P}(A) = \sum_{\omega \in A} p(\omega)$. In a finite uniform space, $\mathbb{P}(A) = |A|/|S|$. Uniformity must come from the model, not merely from drawing a list of possible outcomes.

For a multistep experiment, first specify the rules and the question. Then draw a tree whose levels are successive choices and whose leaves are complete outcomes. Label each edge by the probability of that choice *given its preceding history*; outgoing probabilities sum to one. Multiply along each root-to-leaf path and add the masses of leaves in the desired event. Conditional probability justifies these operations formally.

## 18.2 Monty Hall

Assume a prize is uniform among three doors, the player chooses a door independently and uniformly, and the informed host always opens an unchosen losing door and offers a switch. When the host can choose between two losing doors, assume a fair choice. A complete outcome is $(x, y, z)$: prize location, initial choice, opened door, with $z \neq x, y$. There are twelve outcomes, but they are not equally likely.

If $x = y$, the host has two choices, so each resulting leaf has mass $(1/3)(1/3)(1/2) = 1/18$. If $x \neq y$, the host has one choice, giving mass $1/9$. Switching wins precisely in the latter case: six such leaves give probability $6/9 = 2/3$.

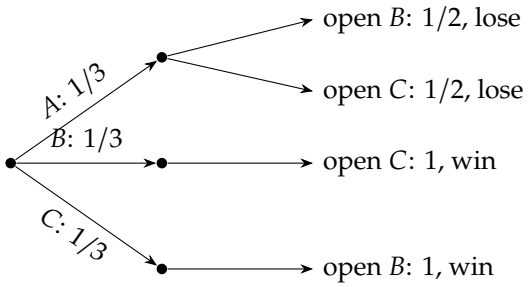



**Figure 4:** A reduced Monty Hall tree, fixing the player's initial door as $A$ by symmetry. Only the prize-at-$A$ branch loses on switching.

Changing the host's rules changes the model. For example, offering a switch only when the original choice wins would make every offered switch lose. The numerical answer therefore depends on the specified procedure.

### 18.3 Intransitive dice

Consider dice whose three equally likely values are $R = \{2, 6, 7\}$, $G = \{1, 5, 9\}$, and $B = \{3, 4, 8\}$. For each pair of independently rolled dice, the nine ordered outcomes are uniform. Direct counting gives

$$\mathbb{P}(R > G) = \mathbb{P}(G > B) = \mathbb{P}(B > R) = \frac{5}{9}.$$

For example, $R$ beats $G$ in $(2, 1), (6, 1), (6, 5), (7, 1), (7, 5)$. Thus pairwise winning probability need not define a transitive order. A player choosing second can select a favorable die against whichever die the first player takes.

Rolling each chosen die twice and comparing sums changes the result. There are $3^4 = 81$ equally likely ordered outcomes. Red's nine ordered two-roll sums are $4, 8, 9, 8, 12, 13, 9, 13, 14$; green's are $2, 6, 10, 6, 10, 14, 10, 14, 18$. Comparing these lists gives red 37 wins, 42 losses, and two ties. Green now wins more often than red. The other comparisons likewise reverse, showing that aggregation can change an intransitive comparison.

**Source.** Z. Abel, B. Chapman, and E. Demaine, *Mathematics for Computer Science*, Lecture 18 (MIT 6.1200J, Spring 2024). Adapted under CC BY-NC-SA 4.0.

# Lecture 19: Conditional Probability

## 19.1 Rules for events

Disjoint events have additive probabilities, including countable disjoint unions. Consequently

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A), \qquad \mathbb{P}(A \setminus B) = \mathbb{P}(A) - \mathbb{P}(A \cap B),$$
$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

Inclusion–exclusion extends to any finite collection of events. Monotonicity gives $A \subseteq B \Rightarrow \mathbb{P}(A) \leq \mathbb{P}(B)$. The union bound, valid without independence, is

$$\mathbb{P}\left(\bigcup_i A_i\right) \leq \sum_i \mathbb{P}(A_i).$$

It also holds for countably many events, by assigning each outcome to the first event containing it and applying disjoint additivity.

## 19.2 Conditioning and products

For $\mathbb{P}(B) > 0$, define

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Conditioning restricts attention to $B$ and renormalizes its mass. Rearranging gives $\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B)$. Iterating this identity yields the chain rule, which explains tree edge probabilities:

$$\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A \mid B \cap C)\mathbb{P}(B \mid C)\mathbb{P}(C),$$

whenever the displayed conditionals are defined. Within a conditioning event, $\mathbb{P}(A \cap B \mid C) = \mathbb{P}(A \mid B \cap C)\mathbb{P}(B \mid C)$.

For a best-of-three series, suppose the first game is fair and each game’s winner wins the next with probability $2/3$. Given that a player wins the first, its series-win probability is

$$\frac{2}{3} + \frac{1}{3}\frac{1}{3} = \frac{7}{9}.$$

The second term follows the win–loss–win route. The probability of having won the first game given a series victory is also $7/9$ in this symmetric example, but reversed conditionals are not generally equal.

## 19.3 Bayes' rule and base rates

Equating the two product-rule expressions for an intersection gives

$$\mathbb{P}(B \mid A) = \frac{\mathbb{P}(A \mid B)\mathbb{P}(B)}{\mathbb{P}(A)}.$$

Here $\mathbb{P}(B)$ is the prior, $\mathbb{P}(A \mid B)$ the likelihood, and $\mathbb{P}(B \mid A)$ the posterior. For hypotheses $B, C$ with positive denominators, the odds form is

$$\frac{\mathbb{P}(B \mid A)}{\mathbb{P}(C \mid A)} = \frac{\mathbb{P}(A \mid B)}{\mathbb{P}(A \mid C)} \frac{\mathbb{P}(B)}{\mathbb{P}(C)}.$$

Choose equally between a fair coin and an always-heads coin. After observing heads, posterior odds of fair to biased are $1 : 2$, so the probability of fair is $1/3$. With prior fair probability $1 - \epsilon$, those odds become $(1 - \epsilon)/(2\epsilon)$: the same observation can lead to a different posterior when the prior changes.

In a hypothetical testing example, prevalence is 0.1, the positive rate among affected people is 0.9, and the positive rate among unaffected people is 0.3. Bayes' rule gives

$$\mathbb{P}(\text{affected} \mid +) = \frac{0.9 \cdot 0.1}{0.9 \cdot 0.1 + 0.3 \cdot 0.9} = \frac{1}{4}.$$

The rates are assumptions of this example. Selecting who gets tested by additional evidence changes the relevant conditioning population.

## 19.4 Aggregating groups and conditioning on evidence

Simpson's paradox is a reversal between within-group and aggregate comparisons. In a hypothetical example, a selective department accepts one of 99 women and zero of one man; another accepts one of one woman and 98 of 99 men. Women have higher acceptance rates in each department, but aggregate rates are 2% for women and 98% for men. There is no algebraic contradiction:

$$\mathbb{P}(A \mid F) = \sum_{d} \mathbb{P}(A \mid F, D = d)\mathbb{P}(D = d \mid F),$$

and the department weights differ between groups.

The same principle applies when evidence is omitted from a question. For an event $G$, background information $A$, and a further observed event $M$, $\mathbb{P}(G \mid A)$ need not resemble $\mathbb{P}(G \mid A \cap M)$. Specify all information being conditioned on before interpreting a probability or likelihood ratio.

**Source.** Z. Abel, B. Chapman, and E. Demaine, *Mathematics for Computer Science*, Lecture 19 (MIT 6.1200J, Spring 2024). Adapted under CC BY-NC-SA 4.0.

# Lecture 20: Independence

## 20.1 Pairwise and mutual independence

Events $A, B$ are INDEPENDENT when $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. If $\mathbb{P}(B) > 0$, this is equivalent to $\mathbb{P}(A \mid B) = \mathbb{P}(A)$. The product definition also handles probability-zero events. Disjoint events of positive probability are dependent, since learning that one occurred excludes the other.

For two independent fair coins, let $A$ mean the first is heads and $B$ mean the two agree. Then $\mathbb{P}(A) = \mathbb{P}(B) = 1/2$ and $\mathbb{P}(A \cap B) = 1/4$, so these events are independent even though their descriptions involve the same coins. With independent coins having head probability $p$, instead

$$\mathbb{P}(A) = p, \quad \mathbb{P}(B) = p^2 + (1-p)^2, \quad \mathbb{P}(A \cap B) = p^2.$$

They are dependent for $0 < p < 1$ except $p = 1/2$; the degenerate cases $p = 0, 1$ are also independent.

A collection $(A_i)_{i=1}^n$ is MUTUALLY INDEPENDENT if

$$\mathbb{P}\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} \mathbb{P}(A_i) \quad \text{for every } I \subseteq [n].$$

Pairwise independence checks only subsets of size two and does not imply mutual independence. For three independent fair coins, let $A, B, C$ respectively mean that coins $1, 2$, coins $2, 3$, and coins $3, 1$ agree. Each event has probability $1/2$ and each pair has intersection probability $1/4$, but $\mathbb{P}(A \cap B \cap C) = 1/4 \neq 1/8$.

Without an independence assumption, marginal probabilities do not determine an intersection. Always $\mathbb{P}(A \cap B \cap C) \leq \min\{\mathbb{P}(A), \mathbb{P}(B), \mathbb{P}(C)\}$; nested events can attain this bound. Multiplying marginals without justification can therefore dramatically understate a joint probability.

## 20.2 Conditional independence

For $\mathbb{P}(C) > 0$, $A, B$ are independent *given* $C$ when

$$\mathbb{P}(A \cap B \mid C) = \mathbb{P}(A \mid C)\mathbb{P}(B \mid C).$$

Conditioning can destroy independence. For independent events $A, B$ each of probability $1/2$, condition on $C = A \cup B$. Then

$$\mathbb{P}(A \mid C) = \mathbb{P}(B \mid C) = \frac{2}{3}, \qquad \mathbb{P}(A \cap B \mid C) = \frac{1}{3} \neq \frac{4}{9}.$$

Dependence after selection does not by itself establish a causal relationship.

## 20.3 Birthday collisions

Let $d \leq n$ independent samples be uniform over $n$ labels. The probability that all labels differ is

$$\prod_{i=0}^{d-1} \left(1 - \frac{i}{n}\right) \leq \exp\left(-\sum_{i=0}^{d-1} \frac{i}{n}\right) = e^{-d(d-1)/(2n)},$$

using $1 - x \leq e^{-x}$. For $d > n$, a collision is certain. The collision threshold is on the order of $\sqrt{n}$, rather than $n$. For the uniform 365-day model, $d = 23$ makes the displayed upper bound on no collision less than $1/2$.

## 20.4 A known model versus an unknown coin

In an independent fair-coin model, past tails do not make heads due; the next head probability remains $1/2$. If the coin's bias is unknown, observations can instead update that model. Suppose prior probability $\epsilon$ selects an always-heads coin and $1 - \epsilon$ selects a fair coin. After $k$ consecutive heads,

$$\mathbb{P}(\text{biased} \mid H^k) = \frac{\epsilon}{\epsilon + (1 - \epsilon)2^{-k}}.$$

The next head probability is this posterior times 1, plus the posterior fair probability times $1/2$. Conditional independence given the coin type need not imply unconditional independence.

**Source.** Z. Abel, B. Chapman, and E. Demaine, *Mathematics for Computer Science*, Lecture 20 (MIT 6.1200J, Spring 2024). Adapted under CC BY-NC-SA 4.0.

# Lecture 21: Random Variables

## 21.1 Functions on outcomes

A random variable is a total function on the sample space. For numerical variables, events such as $\{X = x\}$ and $\{X \geq x\}$ are sets of outcomes whose function values satisfy the condition. The indicator $\mathbf{1}_A$ is one on event $A$ and zero elsewhere. The variable is a function; the event is a subset of outcomes.

For three independent fair coin flips, let $R$ count heads and let $M$ indicate that all three agree. Then $\mathbb{P}(R = 2 \mid M = 1) = 0$, so $R, M$ are dependent. Two discrete variables $X, Y$ are independent when

$$\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y) \quad \text{for every } x, y.$$

Mutual independence of $X_1, \ldots, X_n$ means their joint mass factors for every assignment of all $n$ values. Summing over unused coordinates proves the corresponding condition for every subcollection. This differs from checking only one intersection of $n$ particular events.

For independent fair dice $D_1, D_2$, the sum $S = D_1 + D_2$ depends on $D_1$. Yet $T = \mathbf{1}_{\{S=7\}}$ is independent of $D_1$: for each $d$, exactly one second die gives sum seven, so $\mathbb{P}(T = 1 \mid D_1 = d) = 1/6$. The complementary value $T = 0$ then also has constant conditional probability.

## 21.2 Distributions

The probability mass function is $f_X(x) = \mathbb{P}(X = x)$. The cumulative distribution function is $F_X(x) = \mathbb{P}(X \leq x) = \sum_{y \leq x} f_X(y)$. For any set of possible values $U$, $\mathbb{P}(X \in U) = \sum_{x \in U} f_X(x)$. A Bernoulli variable with success probability $p$ has masses $f(1) = p$, $f(0) = 1 - p$. A uniform variable on $\{1, \ldots, n\}$ has mass $1/n$ at each integer and $F(i) = i/n$ for $1 \leq i \leq n$.

## 21.3 A randomized envelope strategy

An adversary fixes distinct integer amounts $x_0, x_1 \in [0, 100]$. Choose one envelope uniformly, inspect its amount, and decide whether to switch. Always switching or never switching wins with probability $1/2$. An independent random threshold improves this guarantee: choose $Z$ uniformly from $\{0.5, 1.5, \ldots, 99.5\}$; keep the observed amount if it exceeds $Z$, otherwise switch.

If $Z$ lies between the amounts, the strategy always selects the larger. Otherwise its success probability is $1/2$. Exactly $|x_1 - x_0|$ thresholds separate the two integers, so

$$\mathbb{P}(\text{success}) = \frac{1}{2} + \frac{|x_1 - x_0|}{200} \geq \frac{1}{2} + \frac{1}{200}.$$

The guarantee holds for every fixed adversarial choice, with randomness supplied by the algorithm.

## 21.4 The binomial distribution

For $n$ independent Bernoulli trials with common success probability $p$, the number of successes $X$ has distribution $\mathrm{Bin}(n, p)$:

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad 0 \leq k \leq n, \qquad F_X(k) = \sum_{j=0}^{k} \binom{n}{j} p^j (1-p)^{n-j}.$$

The mass formula counts which $k$ trials succeed; all such sequences have the same probability. For fixed $0 < \alpha, p < 1$ with $\alpha n$ integral, Stirling's formula gives the optional approximation

$$\mathbb{P}(X = \alpha n) \sim \frac{1}{\sqrt{2\pi\alpha(1-\alpha)n}} \left[ \left(\frac{p}{\alpha}\right)^{\alpha} \left(\frac{1-p}{1-\alpha}\right)^{1-\alpha} \right]^n.$$

The bracket is at most one, with equality only at $\alpha = p$. For $n = 100, p = 1/2$, the masses at 50 and 25 are approximately $0.0796$ and $1.91 \times 10^{-7}$: values far from the center have very small mass.

**Source.** Z. Abel, B. Chapman, and E. Demaine, *Mathematics for Computer Science*, Lecture 21 (MIT 6.1200J, Spring 2024). Adapted under CC BY-NC-SA 4.0.

# Lecture 22: Expectation

## 22.1 Weighted averages and tail sums

For a discrete numerical random variable $X$, its expectation is

$$\mathbb{E}[X] = \sum_{\omega \in S} X(\omega) p(\omega) = \sum_{x \in \text{range}(X)} x \mathbb{P}(X = x).$$

The second expression groups outcomes with equal values. For signed variables, absolute integrability $\mathbb{E}[|X|] < \infty$ ensures a finite, well-defined mean and justifies rearrangements. Nonnegative variables can have expectation $+\infty$. A fair die has mean $7/2$, although that number is not a possible roll. An indicator satisfies $\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A)$.

For a nonnegative integer-valued variable, another useful expression is

$$\mathbb{E}[X] = \sum_{i=0}^{\infty} \mathbb{P}(X > i) = \sum_{i=1}^{\infty} \mathbb{P}(X \geq i).$$

Indeed, $X = \sum_{i \geq 1} \mathbf{1}_{\{X \geq i\}}$ pointwise; all terms are nonnegative. Equivalently, a mass at integer $j$ is counted once in each of the first $j$ tails. The tail formula has no extra factor $i$, unlike the mass-function formula.

If independent trials fail with probability $p > 0$, the number $T$ through the first failure, including that trial, has $\mathbb{P}(T > i) = (1 - p)^i$. Therefore $\mathbb{E}[T] = \sum_{i \geq 0} (1 - p)^i = 1/p$.

An infinite mean can occur in a valid probability model: let $\mathbb{P}(X = j) = 1/[j(j + 1)]$ for $j \geq 1$. The masses telescope to one, but $\mathbb{E}[X] = \sum_{j \geq 1} 1/(j + 1) = \infty$. Every finite observed sample still has a finite average; finite data alone do not make the underlying mean finite.

## 22.2 Linearity without independence

**Theorem 22.1 [Linearity of expectation].** *For finitely many integrable random variables and fixed constants,*

$$\mathbb{E}\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n} a_i \mathbb{E}[X_i].$$

*Proof.* Expand the definition over outcomes, distribute the finite sum, and interchange the sums: $\sum_\omega p(\omega) \sum_i a_i X_i(\omega) = \sum_i a_i \sum_\omega p(\omega) X_i(\omega)$. No factorization of joint probabilities is used. □

Two dice with fair marginal distributions have expected sum 7 even if their rolls are perfectly dependent. The number of hearts in a uniformly chosen three-card hand is a sum of three position indicators, each of mean $13/52$, so its mean is $3/4$.

### 22.3 Randomly returned phones

Return $n$ phones by a uniformly random permutation. Let $I_i$ indicate that person $i$ receives their own phone, and let $R = \sum_i I_i$. Each $I_i$ has mean $1/n$, hence $\mathbb{E}[R] = 1$. The indicators are not independent, but linearity still applies. For reference, the full distribution is

$$\mathbb{P}(R = k) = \frac{1}{k!} \sum_{j=0}^{n-k} \frac{(-1)^j}{j!}, \qquad 0 \le k \le n.$$

Computing the expectation directly from this distribution is unnecessary. If phones instead undergo a uniformly random cyclic rotation, all $n$ return correctly with probability $1/n$, and none do otherwise. The mean remains 1, despite a very different distribution.

### 22.4 Checking fairness by expectation

Three players ante two candies each and independently guess a fair coin. If all guesses agree, everyone receives their ante back; otherwise the six-candy pot goes to the one or two correct guessers. The first player's net payoffs $0, +1, +4, -2$ occur with probabilities $1/4, 1/4, 1/8, 3/8$, giving mean zero. If the other two players coordinate opposite guesses, the first player can no longer win alone: the payoff is $+1$ or $-2$, each with probability $1/2$, so its mean falls to $-1/2$. Altering joint behavior changes the expected payoff even when individual guesses still look fair.

**Source.** Z. Abel, B. Chapman, and E. Demaine, *Mathematics for Computer Science*, Lecture 22 (MIT 6.1200J, Spring 2024). Adapted under CC BY-NC-SA 4.0.

# Lecture 23: Expectation and Variance

## 23.1 How many events occur?

For events $E_1, \dots, E_n$, let $N = \sum_i \mathbf{1}_{E_i}$ and $\mu = \mathbb{E}[N] = \sum_i \mathbb{P}(E_i)$. Linearity needs no independence. Since $\mathbf{1}_{\{N \geq 1\}} \leq N$, the union bound becomes $\mathbb{P}(N \geq 1) \leq \min\{1, \mu\}$.

Under *mutual independence*, a complementary lower bound is

$$\mathbb{P}(N \geq 1) = 1 - \prod_i (1 - \mathbb{P}(E_i)) \geq 1 - e^{-\mu},$$

using independence of the complements and $1 - x \leq e^{-x}$. This is a mathematical form of Murphy's law: many independent opportunities with a large expected count make at least one occurrence likely. For independent events of probability $1/n$, the exact probability is $1 - (1 - 1/n)^n$, at least $1 - e^{-1}$.

The phone examples show why the assumption matters. For a random rotation, $\mu = 1$ but $\mathbb{P}(N \geq 1) = 1/n$. For a random permutation, inclusion–exclusion on fixed-point events instead gives

$$\mathbb{P}(N \geq 1) = 1 - \sum_{j=0}^{n} \frac{(-1)^j}{j!} \longrightarrow 1 - e^{-1}.$$

The permutation indicators are also dependent; their similar limiting answer does not make the independent-event proof applicable.

As a reliability model, $m$ independent components with common failure probability $p$ all work with probability $(1 - p)^m$. For $m = 130$, this is about $1.13 \times 10^{-6}$ at $p = 0.1$, 0.271 at $p = 0.01$, and 0.878 at $p = 0.001$. Small component-level failure probabilities can accumulate across many components.

## 23.2 Products and ratios

For independent integrable $X, Y$,

$$\mathbb{E}[XY] = \sum_{x,y} xy\mathbb{P}(X = x, Y = y) = \left(\sum_x x\mathbb{P}(X = x)\right)\left(\sum_y y\mathbb{P}(Y = y)\right) = \mathbb{E}[X]\mathbb{E}[Y].$$

The same factorization holds for a finite product of mutually independent integrable variables. Independence is essential: independent fair dice have expected product $(7/2)^2 = 49/4$, whereas a single die squared has mean $(1^2 + \cdots + 6^2)/6 = 91/6$.

There is no general reciprocal or ratio rule. If $X$ is equally likely to be 1 or $-1$, then $\mathbb{E}[1/X] = 0$ while $1/\mathbb{E}[X]$ is undefined. For processor comparisons, averaging workload-specific time ratios and taking a ratio of mean running times measure different quantities. Choose the quantity matching the performance question; an expected ratio need not equal a ratio of expectations, even when all denominators are positive. The ratio itself is a valid random variable when defined.

## 23.3 Variance and moments

For a variable with finite second moment and mean $\mu$,

$$\operatorname{Var}(X) = \mathbb{E}[(X-\mu)^2], \qquad \sigma_X = \sqrt{\operatorname{Var}(X)}.$$

Simply averaging signed deviations gives zero. Squaring prevents cancellation and gives greater weight to large deviations. The $r$th central moment $\mathbb{E}[(X-\mu)^r]$ and absolute central moment $\mathbb{E}[|X-\mu|^r]$ describe other aspects of shape; normalized third and fourth moments give skewness and kurtosis. A normal distribution is determined by its mean and variance, although arbitrary distributions need not be.

For finite $S$, $\sigma_X$ is the weighted Euclidean distance of the value vector from its constant mean vector: $\sigma_X^2 = \sum_\omega p(\omega)(X(\omega)-\mu)^2$. The probability weights are essential in this interpretation. For a symmetric variable taking values $+a, -a$, the mean is zero, variance $a^2$, and standard deviation $|a|$.

## 23.4 Computing and combining variances

Expanding the square and applying linearity gives

$$\operatorname{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2, \qquad \operatorname{Var}(cX+b) = c^2 \operatorname{Var}(X).$$

Thus adding a constant does not change variance; standard deviation scales by $|c|$, including negative $c$. An indicator with success probability $p$ satisfies $I^2 = I$, so $\operatorname{Var}(I) = p - p^2 = p(1-p)$.

For finite-second-moment variables,

$$\operatorname{Var}\left(\sum_i X_i\right) = \sum_i \operatorname{Var}(X_i) + 2\sum_{i<j} \operatorname{Cov}(X_i, X_j), \quad \operatorname{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

Pairwise independence makes each covariance zero and hence makes variances additive. Therefore $H \sim \operatorname{Bin}(n,p)$ has $\mathbb{E}[H] = np$ and $\operatorname{Var}(H) = np(1-p)$. Standard deviations themselves do not add.

In an abstract symmetric-stakes model, split total stake $b$ equally among $k$ independent variables $S_i \in \{-1, 1\}$ with equal probabilities. The combined payoff is $Y = \sum_{i=1}^{k} (b/k) S_i$, with

$$\mathbb{E}[Y] = 0, \qquad \mathrm{Var}(Y) = k(b/k)^2 = \frac{b^2}{k}, \qquad \sigma_Y = \frac{|b|}{\sqrt{k}}.$$

The sum has $k$ terms, one per independent component. Shared dependence introduces covariance terms and can remove this reduction in variance.

**Source.** Z. Abel, B. Chapman, and E. Demaine, *Mathematics for Computer Science*, Lecture 23 transcript (MIT 6.1200J, Spring 2024). Adapted under CC BY-NC-SA 4.0.

# Lecture 24: Large Deviations

## 24.1 Markov's inequality

**Theorem 24.1 [Markov].** *If $X \geq 0$ and $t > 0$, then $\mathbb{P}(X \geq t) \leq \mathbb{E}[X]/t$.*

*Proof.* Pointwise, $X \geq t\mathbf{1}_{\{X \geq t\}}$. Take expectations and divide by the positive number $t$. □

Nonnegativity is necessary: a uniform variable on $\{-1, 1\}$ has mean zero but probability $1/2$ of being at least $1/2$. If $\mathbb{E}[X] > 0$, setting $t = c\mathbb{E}[X]$ gives $\mathbb{P}(X \geq c\mathbb{E}[X]) \leq 1/c$ for $c > 0$.

Known range bounds can improve its use. If scores $X$ lie in $[30, 100]$ with mean 75, applying Markov to $X - 30$ gives $\mathbb{P}(X \geq 90) \leq 45/60 = 3/4$, improving $75/90$. Applying it to $100 - X$ gives $\mathbb{P}(X \leq 65) \leq 25/35$. In general, shift a known lower bound to zero for an upper-tail estimate, or reflect around a known upper bound for a lower-tail estimate.

Markov uses only the mean and can be loose. For correctly returned phones, both random rotation and random permutation have $\mathbb{E}[R] = 1$, so $\mathbb{P}(R \geq n) \leq 1/n$. This is exact for rotation but the permutation's true probability is $1/n!$.

## 24.2 Chebyshev's inequality

**Theorem 24.2 [Chebyshev].** *If $X$ has finite variance, then for every $t > 0$,*

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq t) \leq \frac{\mathrm{Var}(X)}{t^2}.$$

*Proof.* The event is $\{(X - \mathbb{E}[X])^2 \geq t^2\}$. Apply Markov to the nonnegative squared deviation, whose mean is $\mathrm{Var}(X)$. □

For positive standard deviation, the probability of a deviation of at least $c$ standard deviations is at most $1/c^2$. No sign restriction on $X$ is needed. In the score example, variance 25 would give $\mathbb{P}(X \leq 65) \leq \mathbb{P}(|X - 75| \geq 10) \leq 1/4$.

For $R$, the number of heads in $n$ independent fair flips, $\mathbb{E}[R] = n/2$, $\mathrm{Var}(R) = n/4$, and $\sigma_R = \sqrt{n}/2$. For the event $R \geq 3n/4$, Markov gives $2/3$, whereas Chebyshev gives $(n/4)/(n/4)^2 = 4/n$. Pairwise independence already suffices for this variance calculation.

## 24.3 Chernoff's bound

**Theorem 24.3 [Chernoff].** *Let $T_1, \dots, T_n$ be mutually independent variables with $0 \le T_i \le 1$. Write $T = \sum_i T_i$ and $\mu = \mathbb{E}[T]$. For every $c \ge 1$,*

$$\mathbb{P}(T \ge c\mu) \le \exp\bigl[-(c \ln c - c + 1)\mu\bigr].$$

The proof idea is to apply Markov to $c^T$, rather than to $T$ or its squared deviation. For $c > 1$, convexity on $[0, 1]$ gives $c^{T_i} \le 1 + (c-1)T_i$. Independence and $1 + x \le e^x$ therefore imply

$$\mathbb{E}[c^T] = \prod_i \mathbb{E}[c^{T_i}] \le \prod_i \bigl(1 + (c-1)\mathbb{E}[T_i]\bigr) \le e^{(c-1)\mu}.$$

Markov at threshold $c^{c\mu}$ gives the stated exponent. At $c = 1$ the bound is the trivial bound one. Mutual independence, stronger than the pairwise condition used for Chebyshev, justifies the full product.

For fair flips take $c = 3/2$, obtaining

$$\mathbb{P}(R \ge 3n/4) \le e^{-[(3/2)\ln(3/2) - 1/2]n/2} \le e^{-n/20}.$$

This improves a polynomial bound to an exponential one. Taking $c = 1 + 4/\sqrt{n}$ instead bounds $\mathbb{P}(R \ge n/2 + 2\sqrt{n})$ by a quantity tending to $e^{-4} < 0.02$, so it is at most 0.02 for sufficiently large $n$. The typical deviation scale grows like $\sqrt{n}$, while the mean grows like $n$: the proportion of heads concentrates around $1/2$.

**Source.** Z. Abel, B. Chapman, and E. Demaine, *Mathematics for Computer Science*, Lecture 24 (MIT 6.1200J, Spring 2024). Adapted under CC BY-NC-SA 4.0.